

Algebra II with Trigonometry

Chapter 11.1

Study guide

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Study Guide: Circles and Parabolas (Chp 11.1 - Alg 2 Trig/Honors)

1. Short Summary

This document introduces conic sections, focusing specifically on circles and parabolas. It defines both shapes geometrically and algebraically. For circles, it presents their standard equation. For parabolas, it explains the connection to their focus and directrix, as well as giving standard forms for vertical and horizontal axes. The document includes examples for finding key features (focus, directrix, axis) and writing equations for parabolas under given conditions.

2. Core Definitions, Theorems, and Formulas

Conic Sections

- **Conic Sections:** Intersections of a right circular cone and a plane (Circle, Parabola, Ellipse, Hyperbola).

Circles

- **Circle:** Set of all points in a plane equidistant from a fixed point C (the center).
- **Standard Equation (center (h, k) , radius r):**

$$(x - h)^2 + (y - k)^2 = r^2$$

- **Standard Equation (center at origin):**

$$x^2 + y^2 = r^2$$

Parabolas

- **Parabola (Algebraic):** The graph of $y = ax^2 + bx + c$ is a parabola.
- Opens upward if $a > 0$; downward if $a < 0$.
- **Parabola (Geometric):** Set of all points equidistant from a point F (focus) and a line l (directrix).
- **Vertex (V):** Point halfway between focus and directrix.
- **Axis of symmetry:** Line perpendicular to the directrix through the focus.

Standard Forms (Vertex at origin):

- **Vertical Axis:** $x^2 = 4py$
- Focus: $(0, p)$
- Directrix: $y = -p$
- Opens upward if $p > 0$, downward if $p < 0$
- **Horizontal Axis:** $y^2 = 4px$
- Focus: $(p, 0)$
- Directrix: $x = -p$
- Opens right if $p > 0$, left if $p < 0$
- **Latus Rectum:** Line through focus perpendicular to axis; its length is the "focal diameter," which is $|4p|$.

3. Step-by-Step Study Guide

A. Circles

1. **Definition:** All points (x, y) that are a distance r from a center (h, k) .

To Write the Equation:

- Plug the center and radius into $(x - h)^2 + (y - k)^2 = r^2$.
- If the center is the origin, this simplifies to $x^2 + y^2 = r^2$.

B. Parabolas

Algebraic Perspective

- General equation: $y = ax^2 + bx + c$.
- Opens up if $a > 0$, down if $a < 0$.

Geometric Perspective

- **Focus:** Fixed point F .
 - **Directrix:** Fixed line l .
 - **Vertex:** Midpoint between focus and directrix.
 - **Axis:** Perpendicular to the directrix and passes through the focus and vertex.
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Special Case: Vertex at Origin

- **Vertical Axis:**

$$x^2 = 4py$$

- Focus: $(0, p)$, Directrix: $y = -p$
- **Horizontal Axis:**

$$y^2 = 4px$$

- Focus: $(p, 0)$, Directrix: $x = -p$

How to find features: 1. **Given equation**, identify structure ($x^2 = 4py$ or $y^2 = 4px$) to find:

- p : Compare coefficients.

2. **Focus:** Use p . 3. **Directrix:** Set to $-p$ accordingly. 4. **Latus Rectum:** Its length $|4p|$; pass it through focus, perpendicular to axis.

Sketching:

- Draw vertex (origin), axis (y- or x-axis), plot focus and directrix, draw parabola.

4. Practice Problems with Answers

1. Find the equation of a circle centered at (2, -3) with radius 5.

Answer: $(x - 2)^2 + (y + 3)^2 = 25$

2. What is the focus and directrix of the parabola $x^2 = -12y$?

- Compare with $x^2 = 4py \Rightarrow 4p = -12 \Rightarrow p = -3$.
 - **Focus:** $(0, -3)$
 - **Directrix:** $y = 3$
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3. Write the equation of a parabola with vertex at the origin and focus at (4, 0).

- Since focus is on x -axis, it's horizontal: $y^2 = 4px$ with $p = 4$.
 - **Equation:** $y^2 = 16x$
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4. Find the focal diameter of the parabola $y^2 = -20x$

- $4p = -20 \Rightarrow p = -5$
 - **Focal diameter:** $|4p| = 20$
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5. What is the equation of a parabola that opens upward with vertex at origin, and the directrix $y = -6$?

- For a vertical parabola, directrix is $y = -p$. $-p = -6 \Rightarrow p = 6$.
 - **Equation:** $x^2 = 24y$
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5. Assumptions and Ambiguities

- **Assumptions:**
- The equation forms $x^2 = 4py$ and $y^2 = 4px$ always refer to parabolas with the vertex at the origin.

- For all parabolas, axes are either vertical (symmetric about y) or horizontal (symmetric about x), as explicitly stated in the document.
 - When not specified, circle and parabola examples refer to standard position at the origin.
 - **Ambiguities:**
 - Some example problems from the document do not have numerical solutions provided, so for this guide, similar new ones were created.
 - The document presumes knowledge of terms like "latus rectum" and "focal diameter" but defines them only in context; these are incorporated here.
 - The value of p depends on orientation and sign consistent with axis direction as per convention in the document.
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End of Study Guide